No weight that Bombieri–Vinogradov reaches extracts $\sum_{n<N} \Lambda(n)\mu(N-n)$: a no-go for the divisor-switch route, with the exact identities of its design space

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Abstract

Let N be a large even integer and $C(N) = \sum_{n<N} \Lambda(n)\mu(N-n)$. The Huang–Li reduction of binary Goldbach to Elliott–Halberstam [3] consumes its hypothesis through a Möbius-weighted correlation sum in the fixed class $n \equiv N\ (k)$, carrying a weight w_k. Two weights occur: $w_k = 1$, for which the sum is unconditionally small, and $w_k = \log k$, for which it is equivalent to Goldbach itself [6]. Is there a weight between them — one whose divisor sum is still cheap but whose main-term coefficient is still of size 1, which would therefore extract $C(N)$ from Bombieri–Vinogradov alone?

No such weight exists among those whose transform Bombieri–Vinogradov reaches. Writing $b = \mu * w$, the two requirements pull apart: extraction needs mass in b at $K = N^{\theta'}$, while the residual is accessible whenever b is supported below $N^{\theta'-1/2}$, and the thresholds are separated by $N^{1/2}$. Theorem 18 turns that into $\|b\|_1/|B_w| \gg \exp(c_1\sqrt{(\frac{1}{2}+\delta)\log N})$, which exceeds every power of $\log N$ and destroys the extraction *even when the hard input is granted for free*. Theorem 22 shows the separation survives granting $EH(N^{\theta_E})$ for any fixed $\theta_E < 1$, so the obstruction is not a shortage of level. The polynomial family $w_k = f(\log k)$ lies outside the support hypothesis and is closed separately, by a different and weaker argument.

What is not shown. The result is for one evaluation of the switch identity, the triangle inequality; an evaluation exploiting sign correlations among that identity's three terms is not excluded, and nothing here says such correlations are absent. Outside the reached transforms the statement is weaker and is marked as such. The switch identity itself is taken from [6] and not reproved here, so half of Theorem 18 is conditional on it.

The design space is laid out exactly first: six finite rearrangements and one consequence of them identify the per-modulus discrepancy as a dilate of a Möbius–prime correlation, show the signs $\mu(k)$ cancel out of the $\log k$ branch exactly, and exhibit the Goldbach count as a signed sum of nonnegative layers. These are what make the result a statement about a space rather than about two examples. All six are formalised in Lean 4 against Mathlib.

Net progress toward Goldbach: zero. The value of a no-go of this kind is that it closes a design space rather than leaving it to be re-explored.

1 Introduction

1.1 The setting

Let N be a large even integer, $\theta' \in (1/2, 1)$ fixed, and $K = N^{\theta'}$. Put

$$C(N) = \sum_{n<N} \Lambda(n)\mu(N-n), \qquad A(N;k) = \sum_{\substack{n<N \\ n\equiv N\ (k)}} \Lambda(n)\mu(N-n), \tag{1}$$

and $\tilde r(N)=\sum_{n<N}\Lambda(n)\Lambda(N-n)$ for the prime-pair sum, so that $\tilde r(N)>0$ for all large even N is binary Goldbach. For a weight $w:\mathbb{N}\to\mathbb{R}$ put

$$T_w \;=\; \sum_{\substack{k<K\\(k,N)=1}} \mu(k)\,w_k\Big[A(N;k)-\frac{C(N)}{\varphi(k)}\Big]. \tag{2}$$

The functional E_3 of [3] is T_w at $w_k=\log k$; E_4 is not T_w at $w_k=1$ but is obtained from it by partial summation, an inner factor $\log(N-n)$ separating the two [6, Corollary on the collapse]. E_4 is unconditionally $\ll_A N(\log N)^{-A}$, and E_3 is, by an unconditional identity, equivalent to Huang–Li's equation (22) and hence already gives binary Goldbach [6]. This paper closes the part of the space between the two weights that its input reaches, and records what is left open. What is reached is the weights whose transform $b=\mu*w$ is supported below $N^{\theta'-1/2}$; the polynomial family is closed separately and by a weaker argument, and the rest — including $w_k=\log k$ itself — is closed nowhere here. §1.4 says which is which.

Throughout, φ is Euler's function, $\operatorname{rad}(u)=\prod_{p|u}p$, $c,c_1,\dots$ are positive constants depending at most on the parameters each statement fixes (θ', δ, θ_E) and not on N, and

$$\mathfrak{A}(N)=\prod_{q\nmid N}\Big(1-\frac{1}{q(q-1)}\Big),\qquad \mathfrak{S}(N)=2\prod_{p>2}\Big(1-\tfrac{1}{(p-1)^2}\Big)\prod_{\substack{p|N\\p>2}}\Big(1+\tfrac{1}{p-2}\Big). \tag{3}$$

1.2 The design space

Let w be arbitrary and let $b:=\mu*w$, so that $w_k=\sum_{d|k}b_d$; every weight has this form, and b is determined by w. Put

$$B_w:=\sum_{\substack{k<K\\(k,N)=1}}\frac{\mu(k)w_k}{\varphi(k)},\qquad \|b\|_1:=\sum_{u<N}|b_u|. \tag{4}$$

Running the divisor switch of [6] with the weight w gives the identity below. Its derivation is [6]'s and is not repeated here; every negative conclusion of this paper is read off it, so a reader checking this paper alone should treat it as a hypothesis taken from outside. It is not the only one. The others, each used where it is named, are: the bound $\rho_n(x)\ll e^{-c_1\sqrt{\log x}}$ of Huang–Li's Lemma 1, on which Lemma 15 turns; [6]'s residual identity and Bombieri–Vinogradov in the form of [5], both inside Lemma 17; [6]'s theorem on the identity, used in Proposition 12; and Vaughan's $\|S_\Lambda\|_1\gg N^{1/2}$ [7], used in Proposition 27. What is particular to (5) is that it is the one every negative conclusion passes through.

Which of the four is load-bearing is a different question from which is most used, and the answer is not (5). If (5) were wrong, (14) would survive — its proof uses Lemma 15 and not the switch identity — and what would die is the step from (14) to a bound on $|C(N)|$. If Bombieri–Vinogradov reached higher moduli than assumed, the loss factor would shrink but not vanish, and Theorem 22 already measures that. The ρ decay is the one whose failure takes several statements at once: Lemma 15 turns on it, hence (14) and with it Theorem 18, Theorem 22 and Proposition 24(i); and Proposition 12's rate $B_{\log}(K)+\mathfrak{S}(N)\ll e^{-c_1\sqrt{\log K}}\log K$ is the same bound applied to ρ_N, so (13) goes too. That is the single most expensive dependence here, and it is not checked in this packet — Measurement 23 checks Lemma 15 as an identity at $b=\delta_{d_0}$, which tests the algebra and not the decay.

$$B_w\cdot C(N) \;=\; \underbrace{\sum_{u<N}\Lambda(N-u)\mu^2(u)\,b_u}_{\text{complete part}} \;-\; \underbrace{\mathcal{R}_w}_{\text{residual}} \;-\; \underbrace{T_w}_{\text{the object itself}} \;+\;O\Big(N^{o(1)}\Big), \tag{5}$$

the complete part being evaluated by Lemma 2 below, which identifies b as exactly the complete divisor transform. The three named terms are not defined here in the same way: the complete part and T_w are written out above and below, while $\mathcal{R}_w$ is the residual of [6]'s switch, and its formula is not reproduced in this paper. What is used of it is only what Lemma 17 states — that expanding $w_k = \sum_{d|k} b_d$ splits it as $\sum_d b_d \mathcal{R}^{(d)}$ with each $\mathcal{R}^{(d)}$ a sum of Λ over progressions to moduli md. A reader wanting the residual itself has to open [6].

Note 1 (the third term is not a remainder). The term T_w in (5) is (2) itself, and it cannot be absorbed into the $O(N^{o(1)})$: at $w = \log$ it *is* E_3, which [6, Theorem on the identity] evaluates as $\tilde{r}(N) - \mathfrak{S}(N)(N - C(N)) + O_A(N(\log N)^{-A})$. Writing (5) without it turns a rearrangement into a false statement — the two sides then differ by $0.44N$, $0.38N$, $0.32N$ at $N = 2\cdot 10^5$, $4\cdot 10^5$, $8\cdot 10^5$ with $\theta' = 0.56$, which are exactly the measured values of $|T_{\log}|/N$ at those N, read from the run Measurement 7 cites rather than from that measurement's own printed row; over the eight N that run sweeps, the same ratio falls to 0.1245, so over that range the two sides differ by a decreasing fraction of N and not by a constant multiple of it. That the decrease is a power of N is a fit to those eight points and not a bound; the use made of it here is only the lower one, that $|T_{\log}|$ stays above $0.12N$ throughout. The $w = 1$ case of the same collection, $T_1 = \sum H - C(N)B(K)$, is checked numerically and holds to a worst relative error $1.875 \cdot 10^{-16}$ (`lab_weight_gap.py`); (5) itself is not checked numerically anywhere here.

Keeping it explicit is what makes Theorem 18 say something. The identity by itself yields nothing about $C(N)$: an unconditional bound on T_w is exactly what the whole problem asks for, and the only weight for which one is known is $w = 1$, by [6, Theorem on the flat branch]. The theorem below therefore *grants* the bound on T_w and shows the extraction fails anyway.

Lemma 2. *For squarefree u, $\sum_{k|u} \mu(k) w_k = \mu(u)\, b_u$.*

Proof. $\sum_{k|u} \mu(k) \sum_{d|k} b_d = \sum_{d|u} b_d\, \mu(d) \sum_{j|(u/d)} \mu(j) = \sum_{d|u} b_d\, \mu(d)\, [\, u/d = 1\,] = \mu(u) b_u$, using $(d, u/d) = 1$ for squarefree u. □

The two known cases are the two ends of this space: $w = 1$ gives $b = \delta_1$, $\|b\|_1 = 1$ and (Huang–Li's Lemma 1) $B_w \ll e^{-c\sqrt{\log K}}$; while $w = \log$ gives $b = \Lambda$, $\|b\|_1 \asymp N$ — the complete part *is* the binary Goldbach sum — and $B_w \to -\mathfrak{S}(N) \asymp 1$.

1.3 Results

Section 2 records the exact structure of the space: the per-modulus discrepancy is a dilate (Proposition 3), the signs $\mu(k)$ cancel exactly on the $\log k$ branch (Proposition 4), both known weights are the same sum of dilated walls with different weights (Proposition 5), the untruncated object is the Goldbach count itself (Proposition 8), which is therefore a signed sum of nonnegative layers (Proposition 9) over Bombieri–Vinogradov moduli carrying no Möbius (Proposition 10); and the wall cancels out of the count entirely (Proposition 12).

Section 3 proves the no-go: Lemmas 15 and 17 are the two opposing constraints, Theorem 18 is their consequence, and Theorem 22 shows it survives Elliott–Halberstam. Section 4 closes polynomial weights, which lie outside the hypothesis of Theorem 18. Section 5 records the position of the circle method.

1.4 What is not claimed

- **This is a no-go for one method, and for the part of its weight space that method can evaluate.** Theorem 18 closes divisor switching with Bombieri–Vinogradov as the only input, for every weight whose transform that input reaches. Of the weights outside that hypothesis, the polynomial family $w_k = f(\log k)$ is addressed by Proposition 24, whose (ii) is a statement of what is not known rather than an impossibility; the weights that are neither reached by the hypothesis nor polynomial are closed nowhere here. It is not, and does not claim to be, an obstruction to other methods — in particular it says nothing about the spectral large sieve of [4].

- **The genre is classical.** Bombieri's asymptotic sieve [1] shows that sieve weights alone cannot detect primes, however chosen, and the parity obstruction it isolates is the reason one expects a statement of this shape. What is added here is the precise quantitative form for this route (Note 21).

- **Proposition 27 is about two pairings, not about the circle method.** What is shown is that the $(2,2)$ and $(\infty,1)$ Hölder pairings of the integral lie at or above the trivial bound. A circle-method treatment of a binary problem does not ordinarily proceed by a global pairing at all: it splits into major and minor arcs and treats the minor arcs bilinearly. The Parseval floor is taken over the whole circle and so does not depend on where that split is drawn, but a bilinear treatment of the minor arcs is a different mechanism and is not addressed here. Nothing below measures the major arcs or asserts anything about their size.

- **No progress toward Goldbach.** Nothing here bounds $C(N)$, and nothing here is a step toward bounding it.

- **The identities of Section 2 are rearrangements.** Six of the seven are exact and finite, none of those is an estimate, and none makes any quantity smaller. Proposition 12 is the exception: it carries an $O_A(N(\log N)^{-A})$ and uses an asymptotic bound on $|C(N)|$, so it is a consequence of the rearrangements rather than one of them.

2 The design space, exactly

Proposition 3 (the discrepancy is a dilate)**.** *Unconditionally,*

$$A(N;k) \;=\; \mu(k)\,H(N;k), \qquad H(N;k) \;=\; \sum_{\substack{m<N/k \\ (m,k)=1}} \Lambda(N-mk)\,\mu(m). \tag{6}$$

Proof. The class $n \equiv N \pmod{k}$ is $k \mid N-n$, so writing $N-n = mk$ turns the sum into $\sum_{m<N/k} \Lambda(N-mk)\mu(mk)$; and $\mu(mk) = \mu(m)\mu(k)$ when $(m,k)=1$, zero otherwise. □

Identity (6) qualifies the shape of the whole problem. EH_μ is consumed where one Möbius argument *divides* the other, but the per-modulus discrepancy is $\mu(k)$ times a sum in which the two arguments are related by a *difference*, at scale N/k. The two couplings are not two fields; they are the same field seen before and after the switch. It also explains why μ is the noisier of the two inputs in progressions: for μ, $|A(N;k)| = |H(N;k)|$ is a Möbius–prime correlation of length N/k and nothing makes it small, whereas for a random sign pattern ε the sum $\sum_m \Lambda(N-mk)\varepsilon(mk)$ is a sum of independent signs and exhibits square-root cancellation.

Proposition 4 (the weights are nonnegative)**.** *Substituting (6) into $T_{\log}$ and using $\mu(k)^2 = 1$ on squarefree k,*

$$T_{\log} \;=\; \sum_{\substack{k<K \\ (k,N)=1}} \mu^2(k)\,(\log k)\,H(N;k) \;-\; C(N)\,B_{\log}(K), \qquad B_{\log}(K) = \sum_{\substack{k<K \\ (k,N)=1}} \frac{\mu(k)\log k}{\varphi(k)}. \tag{7}$$

Proof. Immediate from (6) and $\mu(k)\mu(k) = \mu^2(k)$. □

The signs $\mu(k)$ cancel exactly: they cancel against the $\mu(k)$ that (6) extracts from the progression sum, and what is left is a sum of dilated Möbius–prime correlations with *nonnegative* weights $\mu^2(k)\log k$. Huang–Li discard the $\mu(k)$ by the triangle inequality; on the $w_k = 1$ branch keeping them is what makes the unconditional theorem of [6] possible; on the $\log k$ branch they are not there to keep. Whatever smallness $T_{\log}$ has must therefore come from the signs of $H(N;k)$ across k. It does not come from there, and the amount is measured. Writing $G = \sum_k (\log k)|H| / \left|\sum_k (\log k)H\right|$ for the gain that cancellation across dilations buys, G reads 1.834, 1.804, 2.207, 2.588 and 2.789 over $N = 2\cdot 10^5$ to $3.2\cdot 10^6$, where the sums run over 313 to 1485 moduli (`lab_positive_weights.py`).

What independent signs would give is not $\sqrt{\#k}$. Writing $x_k = (\log k)H(N;k)$, the quantity G is $\|x\|_1/|\sum_k x_k|$, and under independent signs $|\sum_k x_k|$ is of the size of $\|x\|_2$; so the reference for G is $\|x\|_1/\|x\|_2$. That equals $\sqrt{\#k}$ only when every $|x_k|$ is the same, and Cauchy–Schwarz makes it at most $\sqrt{\#k}$ otherwise. Here the magnitudes are not equal: the same run prints $\|x\|_1/\|x\|_2$ as 11.96, 14.23, 17.90, 21.43 and 26.21, which is 0.66 to 0.69 of the count-based figure. **A count of moduli overstates the reference by about a half.**

Against the reference that is correct for these magnitudes, the measured G of 1.8 to 2.8 is still smaller by an order of magnitude, so the conclusion is unchanged and only the comparison moves. The step from a signed sum to its absolute counterpart — which is what (13) makes below — costs a factor near two on this range, where independence would have bought between twelve and twenty-six.

Proposition 5 (both ends are the same sum of dilated walls)**.** *For any weight w,*

$$T_w \;=\; \sum_{\substack{k<K\\(k,N)=1}} \mu^2(k)\, w_k\, H(N;k) \;-\; B_w\, C(N). \tag{8}$$

In particular

$$T_1 = \sum_{\substack{k<K\\(k,N)=1}} \mu^2(k)H(N;k) - B_1C(N), \qquad T_{\log} = \sum_{\substack{k<K\\(k,N)=1}} \mu^2(k)(\log k)H(N;k) - B_{\log}C(N),$$

and both weights are nonnegative on the range.

Proof. Put (6) into (2). The transform carries $\mu(k)$, so terms with $\mu(k) = 0$ contribute nothing on the left; on the remaining k, $\mu(k)A(N;k) = \mu(k)^2H(N;k)$. The restriction is therefore carried on the right, and the subtracted mean term assembles into $B_wC(N)$ by (4). ☐

Note 6 (the restriction is not decoration)*.* Dropping $\mu^2(k)$ from (8) does not weaken it, it falsifies it: $H(N;k)$ does not vanish at squarefull k, while the left-hand side does. The terms so admitted are of the size of T_w itself — at $\theta' = 0.56$ and $N = 2\cdot 10^5$ they contribute $-0.054316\,N$ at $w = 1$ and $-0.287165\,N$ at $w = \log$, the latter being 0.656087 of $|T_{\log}|$ in absolute value. The fraction is not stable: over $N = 2, 4, 8\cdot 10^5$ it reads 0.656087, 0.701640, 0.788480, so it rises with N and the two thirds is the value at the bottom of that range. It stays above 1/2 throughout, which is the whole of what this note asserts — the restriction is load-bearing and not cosmetic (`audit_squarefull_admitted.py`).

Since $B_1 \ll e^{-c\sqrt{\log K}}$ kills the second term while $B_{\log} \to -\mathfrak{S}(N)$ does not, the unconditional theorem of [6] says exactly that the *flat* sum of dilated walls is $\ll_A N(\log N)^{-A}$, and the Goldbach problem says the same sum weighted by $\log k$ is not small. Everything between what is proved and what is open is the factor $\log k$ inside a positively weighted sum of the same terms.

Measurement 7 (the two ends, measured)**.** Over $N = 2\cdot 10^5$ to $2.56\cdot 10^7$ by doubling, at $\theta' = 0.56$, (8) holds to a worst relative error of $1.875\cdot 10^{-16}$, and

$$\frac{\left|\sum_k \mu^2(k)H(N;k)\right|}{\left|\sum_k \mu^2(k)(\log k)H(N;k)\right|} \;=\; 0.1785,\ 0.1559,\ 0.1484,\ 0.1483,\ 0.1367,\ 0.1334,\ 0.1241,\ 0.1135,$$

falling on average; at the top $|T_1|/N = 0.01425$ against $|T_{\log}|/N = 0.1245$. Fitted against $\log N$ the flat sum decays as $N^{-0.3505}$ and $|T_{\log}|/N$ as $N^{-0.2658}$: the gap widens. Every fitted exponent in this paper is one arithmetic family's. These N are $2 \cdot 10^5 \cdot 2^j$ with $2 \cdot 10^5 = 2^6 \cdot 5^5$, so each has odd radical 5, and the same doubling carries the -0.3620 below and the residual exponent of Measurement 14; `c02_flatsum_k1.txt` states the field for that reason. What these exponents do across radicals is not measured, and no step of the no-go rests on their values. (The neighbouring -0.2696 printed in the same run is the exponent of $|\sum_k (\log k) H(N;k)|/N$, which is a different sum — the two differ by $C(N)B_{\log}(K)$ — and not a second measurement of the same one. Both fall out of one sieve and one index set, so what separates them is the quantity and not the implementation.) The index set here runs from $k = 1$, as (2) and (8) require; note that $H(N;1) = C(N)$ exactly, so dropping that one term removes $C(N)$ from the flat sum — which changes the ratios above by up to 11.6% — taken over the $k \geq 1$ column, which is the one those ratios are formed from; against the $k \geq 2$ column it is 10.4%. Those two percentages are not printed by any run: they are the largest relative gap between the `ratio_code` and `ratio_paper` columns of `c02_flatsum_k1.txt`, attained at $N = 4 \cdot 10^5$ where the columns read 0.1740 and 0.1559, and each is that gap divided by one column in turn. The fitted exponent moves to -0.3620, while leaving $|T_1|$ untouched to 10^{-17}, since the $k = 1$ term of T_1 is $A(N;1) - C(N)/\varphi(1) = 0$. The two columns are computed side by side, from a sieve built independently of the one `lab_weight_gap.py` uses, in `c02_flatsum_k1.py`; that script's loop is the $k \geq 1$ one, and `lab_weight_gap.py`'s starts at $k = 2$. Identity (6) itself is verified over every squarefree $k < N^{0.56}$ coprime to N at $N = 2 \cdot 10^5$, $4 \cdot 10^5$, $8 \cdot 10^5$ with worst relative error $2.294 \cdot 10^{-14}$, $1.197 \cdot 10^{-13}$, $4.591 \cdot 10^{-14}$, and (7) against a direct evaluation of $T_{\log}$ to at most $1.8 \cdot 10^{-16}$ over $N = 2 \cdot 10^5$ to $3.2 \cdot 10^6$ (`lab_weight_gap.py`, `lab_dilate_identity.py`, `lab_positive_weights.py`).

Proposition 8 (the untruncated sum is the count)**.** *Summing over* all k, *with no truncation and no coprimality restriction,*

$$\sum_{k\geq 1} (\log k)\, \mu(k)\, A(N;k) \;=\; -\sum_{n<N} \Lambda(n)\, \mu(N-n)\Lambda(N-n) \;=\; \sum_{p<N} \Lambda(N-p)\log p. \tag{9}$$

So the untruncated object is $\tilde r(N)$, *up to its prime-power part.*

Proof. Both exchanges are finite sums. Since $A(N;k)$ sums n over $k \mid N-n$, exchanging the order and applying $\sum_{d|u} \mu(d)\log d = -\Lambda(u)$ — Möbius inversion of $\log = \mathbf{1} * \Lambda$ — gives the first equality. The second holds because $\mu(u)\Lambda(u) = -\log p$ at $u = p$ prime and vanishes at every higher prime power. $\square$

Proposition 8 is not a decoration: it says that (8) with $w = \log$ is a *rearrangement* of $\tilde r(N)$ rather than an approximation waiting for an estimate, so all of its content is in the truncation at K.

Proposition 9 (the count as signed layers)**.** *Cutting the same double sum along the inner variable of* (6),

$$\sum_k \mu^2(k)(\log k) H(N;k) \;=\; \sum_m \mu(m)\, L(N;m), \qquad L(N;m) := \sum_{\substack{k<N/m,\ (k,m)=1 \\ \mu(k)\neq 0}} (\log k)\, \Lambda(N-mk), \tag{10}$$

and every $L(N;m)$ is nonnegative. With Proposition 8, the Goldbach count is therefore an alternating sum, signed by μ, of nonnegative layers.

Proof. A finite exchange; the coprimality conditions on the pair (k,m) are symmetric. Nonnegativity is that of $\log k$ on $k \geq 1$ and of Λ. The exchange and the nonnegativity are checked numerically in `lab_layer_decomposition.py`, which also measures how much of the total the layers below the truncation carry. $\square$

Proposition 10 (the count over a combined modulus)**.** *Expanding the squarefree condition inside* (10) *by* $\mu^2(k) = \sum_{d^2 | k} \mu(d)$ *and writing* $k = d^2 j$,

$$\sum_{p<N} \Lambda(N-p) \log p \;=\; \sum_{\substack{m,d \text{ squarefree} \\ (d,m)=1,\; md^2<N}} \mu(m)\mu(d)\, L_2(N;m,d), \tag{11}$$

where

$$L_2(N;m,d) = \sum_{\substack{j<N/(md^2) \\ (j,m)=1}} \log(d^2 j)\, \Lambda\big(N - md^2 j\big),$$

and L_2 *carries no squarefree condition. It is not quite a single progression: the surviving* $(j,m) = 1$ *unfolds by Möbius,* $j = ej'$ *with* $e \mid m$*, into progressions to moduli* $md^2 e$*. Only squarefree* e *survive* $\mu(e)$*, so* $e \mid \operatorname{rad}(m)$ *and* L_2 *is a signed sum of at most* $2^{\omega(m)}$ *Bombieri–Vinogradov objects with moduli running up to* $md^2 \operatorname{rad}(m)$ *— equal to* $m^2 d^2$ *exactly when* m *is squarefree. What the squarefree condition bought is that none of them carries a* μ*; what it did not buy is a single modulus.*

Proof. Finite exchange, and $(k,m) = 1$ splits as $(d,m) = (j,m) = 1$. □

Measurement 11 (how large that modulus has to be)**.** Being free of Möbius is a statement about shape, not about level, and the level is the whole question. The truncation below is taken at $md^2 < Q$, while Proposition 10 unfolds that into moduli up to $md^2 \operatorname{rad}(m)$. So the Q^* reported here is a lower bound on the level actually needed — the conservative direction, since what is claimed is that the level must be *large*, and a lower bound that is already large settles that a fortiori. Verified over $N = 2 \cdot 10^5$ to $1.6 \cdot 10^6$ — 166384 to 1331048 pairs — (11) holds to a worst relative error of $4.525 \cdot 10^{-15}$. Truncating it to $q < Q$ and letting $Q^*(N)$ be the least Q past which the deficit stays under εN at the chosen tolerance $\varepsilon = 0.01$,

$$Q^* = 160112,\ 200848,\ 482589,\ 589203, \qquad Q^*/\sqrt{N} = 358.0213,\ 317.5686,\ 539.5509,\ 465.8059,$$

so at that tolerance Q^* exceeds $\sqrt{N}$ by two orders at every N tested. Two limits on that reading, with different grounds, and the difference between them matters.

The first is visible in the four numbers themselves: the ratio is not monotone in N: the second entry is below the first and the fourth is below the third, while the third is above both of them. There is no trend in N here to read, at this tolerance, and none is claimed.

The second is not visible here and is not measured here. The tolerance is chosen and not derived, and *no exponent may be fitted to* Q^*: the deficit oscillates while decaying, so $Q^*(\varepsilon)$ reads the last excursion above εN rather than a trend, and the ratio has no consistent trend in N across tolerances. The sweep establishing that is not in this packet, so the statement about tolerances is reported from the source and not established here.

What stands is therefore a comparison at one fixed tolerance and no more: expanding the squarefree condition buys the Möbius-free moduli and costs two orders in the size of Q^* at $\varepsilon = 0.01$. No rate is claimed, and this statistic does not measure one (`lab_combined_modulus.py`).

Proposition 12 (the wall cancels out of the count)**.** *Substituting* (7) *into the identity of [6, Theorem on the identity], the two occurrences of* $C(N)$ *carry opposite signs and cancel:*

$$\tilde{r}(N) \;=\; \mathfrak{S}(N)N \;+\; \sum_{\substack{k<K \\ (k,N)=1}} \mu^2(k)(\log k)H(N;k) \;-\; C(N)\big(B_{\log}(K) + \mathfrak{S}(N)\big) \;+\; O_A\!\left(\frac{N}{(\log N)^A}\right). \tag{12}$$

Since $B_{\log}(K) \to -\mathfrak{S}(N)$, the quantity $C(N)$ enters the Goldbach count only through the vanishing factor $B_{\log}(K) + \mathfrak{S}(N)$. Consequently

$$\sum_{\substack{k<K \\ (k,N)=1}} \mu^2(k)(\log k)\big|H(N;k)\big| \;\le\; (1-\varepsilon)\,\mathfrak{S}(N)\,N \tag{13}$$

suffices for $\tilde{r}(N) > 0$, for all large even N.

Proof. The limit $B_{\log}(K) \to -\mathfrak{S}(N)$ is the one thing here that (12) does not supply, so it is proved. With $F(s) = \prod_{p\nmid N}(1 - \frac{p^{-s}}{p-1})$ one has $B_{\log}(\infty) = -F'(0)$, and $F(s) = G(s)\zeta(s+1)^{-1}$ with $G(0) = \mathfrak{S}(N)$, so $F'(0) = G(0) = \mathfrak{S}(N)$ and the limit is $-\mathfrak{S}(N)$. The rate is Huang–Li's Lemma 1 in the same form used above, applied to ρ_N and summed by parts against $\log k$ over $k \ge K$, the partial summation being where the extra factor comes from: $B_{\log}(K) + \mathfrak{S}(N) \ll e^{-c_1\sqrt{\log K}}\log K$, which is $o(1)$ uniformly in N once $K = N^{\theta'}$ with θ' fixed.

Given that, the claim is immediate from (12) and $|C(N)| \le \psi(N) \ll N$, the trivial bound, which is all that is needed because the factor multiplying it is the $o(1)$ just bounded rather than $\mathfrak{S}(N)$. □

Note 13 (what the cancellation buys, and what it does not). Condition (13) has the same shape as the constant-factor condition of [6, Proposition on the constant-factor demand]: a bound at level $N^{\theta'}$, with no saving in $\log N$. Nothing has moved off the level axis. What has changed is the threshold constant, from $\mathfrak{S}(N)(1-\mathfrak{A}(N))$ to $\mathfrak{S}(N)$, and the change removes the arithmetic sensitivity along with the slack, because the two were the same thing: $1-\mathfrak{A}(N)$ collapses at primorials, and $\mathfrak{S}(N)$ does not. This is offered as a weaker sufficient condition, proved from two identities. It is not offered as something computation confirms: at accessible N the unspecified O_A term of (12) is numerically the very quantity (13) must bound.

Measurement 14 (the two thresholds, compared)**.** At $\theta' = 0.56$ over $N = 2 \cdot 10^5$ to $3.2 \cdot 10^6$, with $B_H(N) = \sum_{k<K} \mu^2(k)(\log k)|H(N;k)|$, the new ratio $B_H(N)/(\mathfrak{S}(N)N)$ reads 0.4578, 0.4064, 0.4079, 0.3769, 0.3338 — under 1 throughout, and falling overall though not at every step, the third entry sitting above the second — against 2.1519, 1.9105, 1.9175, 1.7720, 1.5691 for the same numerator against the old threshold, $B_H(N)/(\mathfrak{S}(N)(1-\mathfrak{A}(N))N)$, which is above 1 throughout. *The numerator is held fixed across the two lists*; only the threshold constant moves, which is the whole of the comparison. (For the numerator of [6, the constant-factor inequality], which keeps the subtracted mean inside the absolute value and so is a different sum, the old ratio reads 2.1591, 1.9747, 1.9500, 1.7483, 1.5798 — the same conclusion, a different quantity.) Across a seven-point family of arithmetically diverse N the new ratio spans a factor 2.00, where the same family taken against the old threshold with *that* other numerator spans 23.25. The two spans are not formed from the same sum, and the run prints them under column names that say so; what the spans compare is the two thresholds, and the arithmetic sensitivity they differ by belongs to $1-\mathfrak{A}(N)$, which is the only factor between them. The residual coupling is negligible at these N: $|B_{\log} + \mathfrak{S}|/\mathfrak{S}$ stays under 0.0383 and $|C(N)(B_{\log} + \mathfrak{S})|/N$ never exceeds $3.315 \cdot 10^{-4}$, four orders below $\mathfrak{S}(N)$. The residual of (12) decays as $N^{-0.2794}$ with correlation -0.99930 and falls to 2% of N only near $N = 10^{10.16}$ (`lab_direct_route.py`, `lab_direct_identity.py`; the two lists held to one numerator, and the third quantity in the parenthesis, are recomputed side by side in `a1_bh_vs_b.py` — `lab_direct_route.py` prints the parenthetical list and the first of the two above it; only the second, 2.1519, . . . , is exclusive to `a1_bh_vs_b.py`).

3 The no-go

3.1 The two opposing constraints

Lemma 15 (extraction).

$$B_w = \sum_{\substack{d<K\\(d,N)=1}} b_d \frac{\mu(d)}{\varphi(d)}\, \rho_{dN}\left(\frac{K}{d}\right), \qquad \rho_n(x) = \sum_{j<x,\,(j,n)=1} \frac{\mu(j)}{\varphi(j)},$$

and $\rho_n(x) \ll e^{-c_1\sqrt{\log x}}$ whenever $\log n \ll \log x$.

Proof. Write $k = dj$; for squarefree k one has $(d,j)=1$, $\mu(k)=\mu(d)\mu(j)$ and $\varphi(k)=\varphi(d)\varphi(j)$. The bound on ρ is Huang–Li's Lemma 1, a form of Goldston–Yıldırım [2], and the hypothesis $\log n \ll \log x$ is theirs. $\square$

Note 16 (the hypothesis is load-bearing, and it is met where it is used). Without $\log n \ll \log x$ the bound is false: at $\lfloor K/d_0 \rfloor = 1$ the run behind Measurement 23 prints $|B_w|\varphi(d_0) = 1.000000$ over all 5130 admissible d_0 (`audit_extraction_tradeoff.py`; that row is not among the two the measurement reproduces), no saving at all, and that is exactly the corner where $\log n \asymp \log N$ while $\log x = O(1)$. Theorem 18 is unaffected: it maximises only over $d \le N^{\theta'-1/2-\delta}$, where $x = K/d \ge N^{1/2+\delta}$ and $n = dN \le N^{3/2}$, so $\log n \asymp \log x$ holds throughout the range it uses. The same applies to Theorem 22.

Lemma 17 (BV-accessibility). *Expanding $w_k = \sum_{d|k} b_d$ inside the residual gives $\mathcal{R}_w = \sum_d b_d\, \mathcal{R}^{(d)}$, where $\mathcal{R}^{(d)}$ is a sum of Λ over progressions to moduli md with $m < N^{1-\theta'}$. Bombieri–Vinogradov bounds each $\mathcal{R}^{(d)} \ll_A N(\log N)^{-A}$ provided $d \le N^{1/2-\delta}/N^{1-\theta'} = N^{\theta'-1/2-\delta}$; hence, on that support, $\mathcal{R}_w \ll_A \|b\|_1\, N(\log N)^{-A}$.*

Proof. The residual of the switch is [6, the residual identity], which after unfolding μ^2 and the coprimality condition, every term is a sum of Λ over a progression to modulus $q = m\,\mathrm{lcm}(d'^2, e')$ with $m < N^{1-\theta'}$ and auxiliary divisors d', e' at most a fixed power of $\log N$ — written with primes to keep them apart from the d that indexes b_d; Bombieri–Vinogradov is applied in the form of [5]. Inserting the extra divisor $d \mid k$ multiplies the modulus by d, so the level condition $q \le N^{1/2-\delta}$ reads $d \le N^{1/2-\delta}/\big(m\,\mathrm{lcm}(d'^2,e')\big)$; since $\mathrm{lcm}(d'^2,e') \ll (\log N)^{O(1)}$, that factor is absorbed into any $\delta > 0$ and the stated bound on d follows. Summing the $\ll_A N(\log N)^{-A}$ over the support of b with the trivial weight $|b_d|$ gives the claim. $\square$

The two lemmas pull in opposite directions. Lemma 15 says B_w is controlled by the part of b sitting *at* the truncation point K: mass at $d \le K^{1-\varepsilon}$ is damped by $e^{-c\sqrt{\varepsilon \log K}}$. Lemma 17 says the residual is reachable when b lives *below* $N^{\theta'-1/2}$; it does not say that a b living higher puts the residual out of reach, and nothing here does. The two thresholds are separated by a factor $N^{1/2}$.

3.2 The theorem

Theorem 18 (no weight this input reaches extracts $C(N)$). *Fix $\theta' \in (1/2, 1)$ and $0 < \delta < \theta' - \frac{1}{2}$, and let w be any weight whose transform $b = \mu * w$ is supported in $[1, N^{\theta'-1/2-\delta}]$ — the range in which the residual is accessible to Bombieri–Vinogradov. (For $\delta \ge \theta' - \frac{1}{2}$ that range admits at most $b = b_1\delta_1$ — only at $\delta = \theta' - \frac{1}{2}$, and nothing above it — and the statement, though true, is nearly empty.) Then*

$$\frac{\|b\|_1}{|B_w|} \gg \exp\left(c_1\sqrt{(\tfrac{1}{2}+\delta)\log N}\right). \tag{14}$$

Consequently, even granted $T_w \ll_A N(\log N)^{-A}$ *for every* $A > 0$ *— which is what [6, Theorem on the flat branch] supplies unconditionally at* $w = 1$*, and the most that* EH_μ *would supply for a general* w *— identity* (5) *yields no bound on* $|C(N)|$ *better than*

$$|C(N)| \;\le\; \frac{\|b\|_1}{|B_w|} \cdot O_A\!\left(\frac{N}{(\log N)^A}\right),$$

whose prefactor (14) *bounds below by* $\exp(c_1\sqrt{(\frac{1}{2}+\delta)\log N})$*, which exceeds every power of* $\log N$*. So the route delivers no saving of any power of* $\log N$*: no weight satisfying the hypothesis above extracts* $C(N)$ *by divisor switching, not even when the hard input is given for free.*

Two limits on that, both in the statement because they belong there. Weights outside the hypothesis are not covered; Proposition 24 treats the polynomial ones, and by a weaker argument. And what is closed is the absolute-value route, at both of the two places it is taken: $\|b\|_1$ *enters* (14) *by bounding* $\sum_d |b_d|/\varphi(d)\,|\rho_{dN}|$ *termwise, and the three terms on the right of* (5) *are then bounded separately. Neither step is a property of the identity; both are the triangle inequality, and* $\|b\|_1$ *is the budget that inequality spends. An evaluation that uses sign correlations — among the* b_d *in the first step, or among the complete term, the residual and* T_w *in the second — does not factor through either, and is not excluded here. Nothing in this paper says such correlations are absent.*

Proof. By Lemma 15 and $\varphi(d) \ge 1$,

$$|B_w| \le \sum_{d\le D} \frac{|b_d|}{\varphi(d)} \left|\rho_{dN}\!\left(\frac{K}{d}\right)\right| \le \|b\|_1 \max_{d\le D} \left|\rho_{dN}\!\left(\frac{K}{d}\right)\right| \ll \|b\|_1\, e^{-c_1\sqrt{\log(K/D)}},$$

with $D = N^{\theta'-1/2-\delta}$, so that $K/D = N^{1/2+\delta}$. This is (14).

For the second assertion, $\|b\|_1$ and $|B_w|$ are each homogeneous of degree one in w, so their ratio is scale invariant and one may normalise $\|b\|_1 = 1$; that invariance is what makes the normalisation free. It is the scale at which the two thresholds are compared. Solve (5) for $C(N)$ and bound the three terms on the right. The complete part is $\ll \log N$ by Lemma 2 and $\Lambda \ll \log N$; the residual is $\ll_A N(\log N)^{-A}$ by Lemma 17, the support hypothesis on b being exactly what that lemma needs; and $T_w \ll_A N(\log N)^{-A}$ is the granted hypothesis. The last two dominate the first, so

$$|C(N)| \;\le\; \frac{1}{|B_w|} \cdot O_A\!\left(\frac{N}{(\log N)^A}\right) \;=\; \frac{\|b\|_1}{|B_w|} \cdot O_A\!\left(\frac{N}{(\log N)^A}\right).$$

This is the bound the route produces, and it is the displayed one. What (14) adds is a lower bound on its prefactor: the prefactor is $\gg \exp(c_1\sqrt{(\frac{1}{2}+\delta)\log N})$, which exceeds every power of $\log N$, so no choice of w in the design space turns the displayed bound into a saving of a power of $\log N$. The inequality is not run in the other direction — (14) bounds the prefactor below and so cannot be substituted into an upper bound for $|C(N)|$; what it establishes is that the one evaluation this route offers — the triangle inequality applied to (5) with the bound on T_w granted — gives nothing better than the stated ratio. It says nothing about evaluations that do not factor through that inequality. □

Note 19 (why granting T_w is the right form). Without the granted hypothesis the theorem would be vacuous rather than strong: (5) could then be satisfied by T_w alone being large, and it says nothing about $C(N)$ at all. Granting it isolates what the design space actually fails at. The failure is not a shortage of input — the input is handed over — but the arithmetic of two thresholds that lie a factor $N^{1/2}$ apart. That is why raising the level does not help either (Theorem 22).

The loss factor is $\exp(c_1\sqrt{(\frac{1}{2}+\delta)\log N})$, and the $1/2$ in the exponent is literally the exponent of the $\sqrt{N}$ barrier: it is the gap between where a weight must live to see $C(N)$ (at $N^{\theta'}$, the truncation point) and where it may live for Bombieri–Vinogradov to close its residual (below $N^{\theta'-1/2}$). The two known weights exhibit the two endpoints of this space; Theorem 18 shows the interior is empty where its hypothesis reaches, and shows why.

Note 20 (which branch the no-go covers). The two endpoints are not on the same side of Theorem 18's support hypothesis, and the asymmetry is worth stating because it runs opposite to the way the two branches are usually named. That hypothesis asks the transform $b = \mu * w$ to be supported in $[1, N^{\theta'-1/2-\delta}]$, and

$$w_k = 1 \implies b = \mu * 1 = \delta, \qquad w_k = \log k \implies b = \mu * \log = \Lambda,$$

verified elementwise to $d \le 4000$. The delta is inside the hypothesis for any positive bound; the support of Λ runs to the truncation point, measured at 1369 and 2971 against bounds 2.2 and 2.4 at $\theta' = 0.56$. Those bounds are $N^{\theta'-1/2}$, the $\delta = 0$ endpoint, which is the largest the hypothesis ever allows; any $\delta > 0$ makes them smaller and the gap wider, so the comparison is stated in the direction least favourable to it. So Theorem 18 covers the *flat* branch and not the log one — the flat branch being the one [6, Theorem on the flat branch] settles unconditionally, and the log branch the one left open. That is consistent rather than circular: what is forbidden here is *extracting* $C(N)$ from a weighting, and [6, Theorem on the flat branch] does not extract $C(N)$; it shows the weighting is small, which is a different thing.

Below the threshold the statement is empty rather than false. For $\theta' < 1/2$ the support bound $N^{\theta'-1/2-\delta}$ is under 1 — the run prints 0.5 and 0.4 at $\theta' = 0.44$ for $\delta = 0$, which is the largest the bound can be — so no weight meets the hypothesis at all, the delta included. That Theorem 18 says nothing there is not a gap in it: it is the same Bombieri–Vinogradov reach that governs $\theta' > 1/2$ throughout, appearing here in the theorem's own hypothesis (`audit_nogo_scope.py`).

Note 21 (relation to Bombieri's asymptotic sieve). The genre of Theorem 18 is classical. Bombieri's asymptotic sieve [1] shows that sieve weights alone cannot detect primes, however the weights are chosen, and the parity obstruction it isolates is the reason one expects a statement of this shape. What Theorem 18 adds is not the phenomenon but its precise form here: the obstruction is quantified as a separation of $N^{1/2}$ between two thresholds, it is specific to the divisor-switch route, and it survives granting the full Elliott–Halberstam conjecture rather than merely Bombieri–Vinogradov. The reason the statement is worth making is narrow but real: without it, the $\log k$ weight looks like one member of a family of candidate consumptions, and the identity of [6, Theorem on the identity] then looks like an accident of one choice of weight rather than a property of the route.

3.3 The no-go survives Elliott–Halberstam

The obvious objection to Theorem 18 is that Bombieri–Vinogradov is not the only available input: Huang–Li's Theorem 1 itself *assumes EH* for Λ at level N^{θ}. Raising the assumed level does not help.

Theorem 22. *Suppose Λ has level of distribution $\theta_E \in (0,1)$, i.e. $EH(N^{\theta_E})$ holds. Then the residual is accessible only for b supported on $d \le N^{\theta_E-(1-\theta')}$, while extraction still requires mass at $d \asymp K = N^{\theta'}$; the two thresholds are separated by $N^{1-\theta_E}$, and*

$$\frac{\|b\|_1}{|B_w|} \gg \exp\Big(c_1\sqrt{(1-\theta_E)\log N}\Big),$$

which exceeds every power of $\log N$ for each fixed $\theta_E < 1$. On $\theta_E \le 1-\theta'$ the support condition admits no $d \ge 1$ and the statement is vacuous, so what is established is the range $\theta_E \in (1-\theta', 1)$.

Proof. Identical to Theorem 18 with $N^{1/2-\delta}$ replaced by N^{θ_E}: the residual's moduli are md with $m < N^{1-\theta'}$, so $d \le N^{\theta_E}/N^{1-\theta'}$, whence $K/D = N^{\theta'}/N^{\theta_E-1+\theta'} = N^{1-\theta_E}$, and Lemma 15 gives the ratio.

This asks $\theta_E > 1-\theta'$. Below that the support bound $N^{\theta_E-(1-\theta')}$ is under 1, so no $d \ge 1$ meets it and the hypothesis on b is empty — in the same way Note 20 records for Theorem 18. The displayed exponent is therefore established on $\theta_E > 1-\theta'$ and the statement is vacuous below; for $\theta' > 1/2$ the region left out is $\theta_E < 1/2$, beneath what Bombieri–Vinogradov already gives. □

Closing the gap would require $\theta_E = 1$ exactly: equidistribution of Λ in progressions to moduli of size N itself, where each progression contains $O(1)$ terms and the statement carries no information. So the route stays closed *even granting the full Elliott–Halberstam conjecture* — it needs not a stronger level but a different mechanism.

Measurement 23 (the load-bearing lemma, checked as an identity)**.** Lemma 15 carries the theorem, so it is checked directly at $N = 99\,999\,998$, $\theta' = 0.56$, $K = \lfloor N^{\theta'} \rfloor = 30199$. For the single-divisor weights $w_k = [\,d_0 \mid k\,]$, i.e. $b = \delta_{d_0}$, the lemma is an identity and is verified as one: over all $10\,262$ squarefree $d_0 < K$ coprime to N, the brute-force B_w over $k < K$ and the factorised $\mu(d_0)\varphi(d_0)^{-1}\rho_{d_0N}(K/d_0)$ agree to $6.505 \cdot 10^{-18}$, the worst case being $d_0 = 33$.

The size behaves as the lemma predicts. The quantity $|B_w|\varphi(d_0)$ is of order 1 only when $K/d_0 = O(1)$, i.e. only when the weight's transform sits at the truncation point, and is damped throughout the range Bombieri–Vinogradov admits: for these parameters $N^{\theta'-1/2} = 3$, and over $d_0 \le 3$ the largest value attained is 0.004145. The value is not a function of K/d_0 alone — ρ_{d_0N} carries the condition $(j, d_0N) = 1$, so it depends on which small primes divide d_0. At $\lfloor K/d_0 \rfloor = 7$ the 184 admissible d_0 produce exactly four values, split by $\gcd(d_0, 15)$:

$\gcd(d_0, 15)$	1	3	5	15
$\lvert B_w\rvert\,\varphi(d_0)$	0.250000	0.750000	0.500000	1.000000

and at $\lfloor K/d_0 \rfloor = 13$ the 55 admissible d_0 produce nine values spanning 0.066667 to 0.816667, the largest at the d_0 divisible by 15 and by no other prime below 14, where $j = 3$ and $j = 5$ drop out of ρ_{d_0N} and what is left is $1 - \frac{1}{10} - \frac{1}{12}$ (`audit_extraction_tradeoff.py`). The list stood here as six values to 0.566667 until the run behind it was found to be printing only the first six of its own column, without saying so; the three it dropped — 0.650000, 0.666667, 0.816667 — were the three largest. So the decay in $\lfloor K/d_0 \rfloor$ is slower than this measurement was read as saying — which does not touch Theorem 18, whose range is $d_0 \le 3$, i.e. $\lfloor K/d_0 \rfloor \ge 10066$, where the maximum is 0.004145.

4 Smooth weights: $\mu * \log^D = \Lambda_D$

Theorem 18 closes the weights whose transform $b = \mu * w$ is supported low enough for Bombieri–Vinogradov. That hypothesis excludes the most natural family of all, $w_k = f(\log k)$ with f a polynomial, whose transform is spread over every scale. For those the structure is explicit:

$$b \;=\; \mu * \log^D \;=\; \Lambda_D,$$

the generalised von Mangoldt function. Λ_D vanishes on integers with more than D prime factors, and on a squarefree $u = p_1 \cdots p_r$ with $r \le D$ it is the r-th finite difference of x^D at the points $\log p_i$; in particular $\Lambda_D(u) = D!\,\log p_1 \cdots \log p_D$ when $r = D$. Hence for $f = x^D$ the complete part splits by the number of prime factors,

$$\mathrm{CP}_D(N) = \sum_{r=1}^{D} \sum_{\substack{u<N \text{ squarefree} \\ \omega(u)=r}} \Lambda(N-u)\,\Lambda_D(u), \tag{15}$$

the $r = 1$ piece being a Goldbach-type sum and the $r \geq 2$ pieces Chen-type sums (prime plus r-almost-prime).

Proposition 24 (polynomial weights)**.** *Let $w_k = f(\log k)$ with $f = \sum_{j \leq D} c_j x^j$ real.*

(i) If $D = 0$ then $B_w \ll e^{-c\sqrt{\log K}}$ (Huang–Li's Lemma 1): no extraction.

(ii) If f is a single monomial x^D, $D \geq 1$, then every term of CP_D is nonnegative, since $\Lambda \geq 0$ and $\Lambda_D \geq 0$; so $\mathrm{CP}_D \geq 0$ and no cancellation is available inside it. *Classical sieve upper bounds are expected to give $\mathrm{CP}_D \ll N(\log N)^{D-1}$; that bound is quoted as standard, is not proved here, and nothing below rests on it. A matching lower bound is not available: at $D = 1$ the sum is exactly the binary Goldbach sum $\sum_{p<N} \Lambda(N-p)\log p$, whose order is the assertion to be proved. So CP_D is not known to be $o(N)$, and at $D = 1$ deciding whether it is would settle Goldbach.*

(iii) Cancellation therefore requires coefficients of opposite sign, i.e. tuning c_1 against $c_2, \ldots$; but the ratio that would have to be matched involves the asymptotics of the $r = 1$ pieces $\sum_{p<N} \Lambda(N-p)\log^j p$, whose order is unknown for every $j \geq 1$ and which at $j = 1$ is *the binary Goldbach sum. The tuning is circular.*

Consequently no polynomial weight makes the complete part a classically known object.

Proof. (i) is Huang–Li's Lemma 1 applied to $w \equiv 1$. For (ii), $\Lambda_D \geq 0$ on the squarefree integers with $\omega \leq D$ because for $u = p_1 \cdots p_r$ the value $\Lambda_D(u)$ is the r-fold mixed difference of $x \mapsto x^D$ at $\log p_1 < \cdots < \log p_r$, and an r-fold difference of a function whose r-th derivative $D(D-1)\cdots(D-r+1)x^{D-r}$ is nonnegative is itself nonnegative; convexity alone would give this only for $r \leq 2$. Every summand of (15) is then a product of nonnegative factors, so no cancellation is available and $\mathrm{CP}_D \geq 0$. The upper bound quoted in the statement is the classical sieve upper bound for prime-plus-almost-prime sums applied to each r-piece; it is not proved here and (ii) does not use it — what closes (ii) is nonnegativity together with the absence of a matching lower bound. No lower bound of the same order is claimed: the $r = 1$ piece is a prime-plus-prime count, on which the sieve gives none — that is the parity obstruction of Note 21 — and at $D = 1$ it is the whole of CP_1. For (iii), a degree-D tuning has D free coefficients and the constraint that must be solved is an identity between the leading asymptotics of the r-pieces; the $r = 1$ piece with $D = 1$ is $\sum_{p<N} \Lambda(N-p)\log p$, whose asymptotic is the assertion to be proved. □

Note 25 (the canonical tuning moves the wrong way). Since $b = \mu * w$ we have $w = \mathbf{1} * b$ and hence $B(s) = W(s)/\zeta(s)$. That series has a *double* pole at $s = 1$ for every degree-2 f, so no such f removes it and b cannot be made mean-zero; the most a degree-2 tuning can do is kill the second-order term. With $f = x^2 + cx$ one has $W = \zeta'' - c\zeta'$, and neither ζ'' nor ζ' carries a $(s-1)^{-1}$ term.

Which way it moves is measured, and the title is not rhetorical. At $N = 10^6$ the untuned CP is $39.8793\,N$; the canonical tuning $x^2 + (2+2\gamma)x$ *raises* it to $45.4201\,N$, a rise of 13.89 per cent, while $x^2 - 2\gamma x$ lowers it to $37.8516\,N$, by 5.08 per cent. The same ordering holds at $4 \cdot 10^6$ and $1.6 \cdot 10^7$ (`audit_polyweight.py`). Neither tuning buys smallness, and the canonical one is the worse of the two: killing the mean of b is not what is wanted. What must be small is b's correlation with $\Lambda(N - \cdot)$, not its mean.

Measurement 26 (the two pieces are the same order)**.** Nonnegativity, and not a separation of scales, is what closes (ii). At $N = 10^6$, $4 \cdot 10^6$, $1.6 \cdot 10^7$ the two pieces of CP_2 are the same order:

N	10^6	$4 \cdot 10^6$	$1.6 \cdot 10^7$
$r = 1:\ \sum_p \Lambda(N-p)\log^2 p \,/\, N$	22.5145	25.0438	27.4579
$r = 2:\ 2\sum_{pq} \Lambda(N-pq)\log p \log q \,/\, N$	17.3648	19.7926	22.2512
ratio r_2/r_1	0.7713	0.7903	0.8104
$\mathrm{CP}_2/(N\log N)$	2.8866	2.9494	2.9967

The ratio drifts toward 1, not toward 0 or ∞. Calibration: the $D = 1$ column reproduces the Goldbach sum $\sum_p \Lambda(N-p)\log p = 1.7565N$, $1.7633N$, $1.7614N$ against $\mathfrak{S}(N) = 1.76043$, which is the same at all three N because N has the same prime support $\{2, 5\}$ in each case (`audit_polyweight.py`).

5 Both standard pairings have zero margin on $C(N)$

Since the switch is closed over the part of its design space Bombieri–Vinogradov evaluates, and of the remainder only the polynomial weights are reached, in the weaker sense of Proposition 24, it is worth recording the exact position of the other classical mechanism. Write

$$C(N) = \int_0^1 S_\Lambda(\alpha)\, S_\mu(\alpha)\, e(-N\alpha)\, d\alpha,$$

where

$$S_\Lambda(\alpha) = \sum_{n\le N} \Lambda(n) e(n\alpha), \qquad S_\mu(\alpha) = \sum_{m\le N} \mu(m) e(m\alpha).$$

Proposition 27 (zero margin)**.** *The two pairings a circle-method estimate of this integral normally uses —* $(2,2)$ *and* $(\infty,1)$ *— lie at or above the trivial bound* $|C(N)| \le \psi(N) \sim N$*:*

(i) $\|S_\Lambda\|_2 \|S_\mu\|_2 = \big(\sum_{n\le N}\Lambda(n)^2\big)^{1/2}\big(\sum_{m\le N}\mu^2(m)\big)^{1/2} \sim (6/\pi^2)^{1/2} N(\log N)^{1/2}$*, exceeding the trivial bound by a factor* $\asymp (\log N)^{1/2}$*, which* grows*;*

(ii) any bound of the shape $\sup_\alpha |S_\mu(\alpha)| \cdot \|S_\Lambda\|_1$ *is at least* $\|S_\mu\|_2 \|S_\Lambda\|_1 \gg N^{1/2}\cdot N^{1/2} = N$*.*

Proof. (i) is Cauchy–Schwarz followed by $\sum_{n\le N}\Lambda(n)^2 \sim N\log N$ and $\sum_{m\le N}\mu^2(m) \sim 6N/\pi^2$. For (ii), the two factors are bounded below for different reasons. Parseval gives $\sup_\alpha |S_\mu| \ge \|S_\mu\|_2 = (6N/\pi^2)^{1/2}$, an identity-level constraint: no improvement in Möbius exponential-sum technology can lower $\sup_\alpha |S_\mu|$ below $(6/\pi^2)^{1/2}N^{1/2}$. For the second factor Parseval runs the *wrong way*: it gives $\|S_\Lambda\|_1 \le \|S_\Lambda\|_2 \ll (N\log N)^{1/2}$, an upper bound, while the trivial lower bound $\|S_\Lambda\|_1 \ge \big|\int_0^1 S_\Lambda(\alpha)e(-n\alpha)d\alpha\big| = \Lambda(n)$ reaches only $\log N$. The bound $\|S_\Lambda\|_1 \gg N^{1/2}$ that (ii) uses is a theorem of Vaughan [7]. □

The classical uniform bound $S_\mu(\alpha) \ll_A N(\log N)^{-A}$, due to Davenport and quoted here as standard rather than proved, is therefore useless here: it is a saving over N, whereas the pairing needs a bound at the scale $N^{1/2}$, which Parseval forbids. This is the parity obstruction in circle-method language — *the binary problem sits exactly at the trivial bound with no margin* — and it is why the method that settles the ternary problem cannot be pushed to the binary one by any sharpening of the Möbius input.

Measurement 28 (the constants)**.** Computed by exact FFT on a $4N$-point grid at $N = 2^{14}, \ldots, 2^{20}$:

N	2^{14}	2^{16}	2^{18}	2^{20}
$\|S_\mu\|_2/\sqrt{N}$	0.7798	0.7797	0.7797	0.7797
$\sup\|S_\mu\|/\sqrt{N}$	3.058	2.742	2.853	2.801
$\|S_\Lambda\|_1/\sqrt{N}$	1.946	2.084	2.219	2.346
(i) bound$/N$	2.297	2.473	2.639	2.795
$N/\big(\sup\|S_\mu\|\,\|S_\Lambda\|_1\big)$	0.168	0.175	0.158	0.152

The first row reproduces $\sqrt{6/\pi^2} = 0.779697$ to four significant figures — the entries before rounding are 0.779764, 0.779725, 0.779686, 0.779699 — confirming the computation; the last row — the margin that route (ii) would need to exceed 1 — sits near 0.16 and falls over the last three abscissae. Route (i) diverges from the trivial bound like $(\log N)^{1/2}$, as predicted. For reference the object itself is small: $C(N)/N = -0.0105, 0.0001, 0.0059, 0.0032$ at these N (`audit_circle_margin.py`).

6 Summary

- The design space of divisor-switch weights is described exactly by six finite rearrangements — Propositions 3, 4, 5, 8, 9 and 10 — with Proposition 12 a consequence of them rather than one of them: the per-modulus discrepancy is a dilate, the signs $\mu(k)$ cancel on the $\log k$ branch, and the untruncated object is the Goldbach count itself.

- Theorem 18: no weight whose transform is Bombieri–Vinogradov-accessible extracts $C(N)$ by the evaluation this route offers, and this holds even when the bound on the switch identity's third term T_w — the object the whole problem is about — is granted for free. What is closed is the triangle-inequality evaluation; an evaluation through sign correlations among the identity's three terms is not. The loss is $\exp(c_1\sqrt{(\frac{1}{2}+\delta)\log N})$, and the 1/2 is the exponent of the square-root barrier.

- Theorem 22: the separation survives $EH(N^{\theta_E})$ for every fixed $\theta_E < 1$ at which the hypothesis is non-empty, that is on $\theta_E \in (1-\theta', 1)$. Closing it would require $\theta_E = 1$, where the statement is vacuous.

- Proposition 24: polynomial weights, which lie outside the hypothesis of Theorem 18, are met by nonnegativity of Λ_D and by circularity of the tuning — which leaves their complete part not known to be $o(N)$ rather than shown to be large. That is a statement about what is not known and not an impossibility, and the weights that are neither reached by the hypothesis nor polynomial are closed nowhere here.

- Proposition 27: the two pairings a circle-method estimate of $C(N)$ normally uses have zero margin, with Parseval supplying the floor. A bilinear treatment of the minor arcs is a different mechanism and is not addressed.

- Net progress toward Goldbach: zero. What is closed is a design space, so that it need not be re-explored.

References

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[6] Y. Lee, *The Huang–Li reduction of Goldbach to Elliott–Halberstam does not decompose, with an unconditional bound for a Möbius-weighted correlation sum in fixed residue classes*, 2026. Companion paper, in the directory `P1-mobius-fixed-class/` of the repository named in Appendix B, with its own code and output. The switch identity (5) is its Theorem C and is not reproved here.

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Disclosure statement

No potential competing interest, financial or non-financial, was reported by the author. No funding was received for this work. An AI assistant was used in preparing it; what tool, how, and why is set out in the appendix, together with what was machine-verified, what was checked by computation, and what was neither.

Data availability statement

The code behind every figure printed here, the output it produced, and the Lean 4 development are openly available at `https://github.com/soke91/arithmetic-convolution-fields`, in the directory `P2-no-go-divisor-switch/`. Two percentages quoted in Measurement 7 are read off two printed columns of a cited result file rather than produced by a run, and that passage says which columns and at which N. No data derived from human or animal subjects were used; every quantity here is computed from the integers.

A Use of an AI assistant, and what checked what

This work was done in close collaboration with an AI assistant, which carried much of the derivation and all of the computation. The author takes full responsibility for every claim made here.

Which tool, and how. The manuscript in its present form was prepared with Anthropic's Claude, model Opus 5, run through the Claude Code command-line interface. It was used to derive and check the identities, to write and run every script in the accompanying code, to write the Lean development, and to draft and revise this text. Earlier stages used earlier models of the same family; the repository does not record which, and nothing is relied on from those stages that has not since been rederived and rerun.

What follows says what was checked and by what, rather than asserting that it was checked.

Machine-verified. The finite rearrangements that lay out the design space are formalised in Lean 4 against Mathlib, in the file `lean/GoldbachLean/Layers.lean` of the code packet: the exchange of the double sum along its inner variable that Proposition 9 performs, the nonnegativity of the layers it produces, and the sense in which the two ends of Proposition 5 are one expression. Each depends on `propext`, `Classical.choice` and `Quot.sound` and on nothing else; the axiom report is in the packet. The Lean statements take Λ, μ and log as arbitrary functions, so what is verified is that these identities use no arithmetic property of them — which is what makes them rearrangements rather than estimates.

The no-go itself is not machine-verified, and neither theorem of this paper has had a reading by a subject expert. Theorem 18 rests on Bombieri–Vinogradov, which Mathlib does not carry, so no part of it is formalised. What is formalised is the scaffolding, not the conclusion.

Checked by computation. Every figure quoted is produced by a named run in the accompanying code, with the two exceptions named in Measurement 7, which are read off two printed columns of a cited file. A check in the build refuses a printed figure that no cited run produces. Forty-three rules carry a verdict across the fourteen cited runs, and every one is reported with it, including the ten the runs did not meet.

Cross-read. The proofs and the computations were additionally read by separate AI sessions working without sight of each other's findings, and several statements here are corrections those readings produced — among them the reference against which the cross-k gain is compared, which an earlier printing took to be a count of moduli. It is not a substitute for either of the checks named as absent above.

B Reproducibility

The code and the output it produced are at

`https://github.com/soke91/arithmetic-convolution-fields`

in the directory `P2-no-go-divisor-switch/`. Each entry below is run as `python code/<script>.py` from that directory and writes the file of the same name in `results/`. The figures here were produced under Python 3.12.10 and NumPy 2.2.5. That address is mutable, so a number checked against the branch head may have been recomputed since; the release named there pins the state these figures were read from.

Two files are in that directory without being named below: `lab_layer_tail.py`, whose output `lab_combined_modulus.py` reads, and `indep.py`, which `c02_flatsum_k1.py` imports. Run `lab_layer_tail` before `lab_combined_modulus`; the rest are independent of each other.

A nonzero exit status is not a failure. These runs carry preregistered rules, and a run whose own rule was refuted says so in its exit code as well as in its output. Eight of the fourteen runs named below record such a refutation, and all eight are named in §1.4.

Script	Result file	Supports
`lab_dilate_identity.py`	`lab_dilate_identity.txt`	Proposition 3
`lab_positive_weights.py`	`lab_positive_weights.txt`	Proposition 4
`lab_weight_gap.py`	`lab_weight_gap.txt`	Measurement 7, $k \geq 2$ column
`c02_flatsum_k1.py`	`c02_flatsum_k1.txt`	Measurement 7, $k \geq 1$ column
`lab_direct_identity.py`	`lab_direct_identity.txt`	Proposition 8
`lab_layer_decomposition.py`	`lab_layer_decomposition.txt`	Proposition 9
`lab_combined_modulus.py`	`lab_combined_modulus.txt`	Proposition 10
`lab_direct_route.py`	`lab_direct_route.txt`	Measurement 14, residual and parenthetical list
`a1_bh_vs_b.py`	`a1_bh_vs_b.txt`	Measurement 14, the two lists
`audit_extraction_tradeoff.py`	`audit_extraction_tradeoff.txt`	Measurement 23
`audit_polyweight.py`	`audit_polyweight.txt`	Measurement 26
`audit_circle_margin.py`	`audit_circle_margin.txt`	Measurement 28
`audit_nogo_scope.py`	`audit_nogo_scope.txt`	Note 20
`audit_squarefull_admitted.py`	`audit_squarefull_admitted.txt`	the note after Proposition 5

No computation is used as a premise of any proof. Six of the seven statements of Section 2 are finite rearrangements and are verified as such to machine precision; Proposition 12 is the seventh and is not one of them — it carries an $O_A(N(\log N)^{-A})$ — so what is verified there is its identity part and not the error term. The measurements record sizes over the ranges stated in each statement.

Eight of the fourteen runs record a pre-registered rule their own output did not meet, and this paper reported none of them by name. They are: `audit_extraction_tradeoff` (two rules, one of them the damping claim itself — above $\lfloor K/d_0 \rfloor = 1000$ the maximum is 0.062792 against a registered cap of 0.05; the range Theorem 18 uses is $\lfloor K/d_0 \rfloor \geq 10066$, where the maximum is 0.004145, so the theorem is untouched and the registered cap was set too tight); `lab_combined_modulus` (two rules), whose control with the sign of $\mu(d)$ randomised has a *smaller* median deficit than μ at three of its four N, including the two largest — 0.0454 against 0.0547 and 0.0342 against 0.0733 — and a smaller minimum at every one, so that measurement does not separate μ from a sign pattern at all in the range it covers, and what makes the far pairs negligible is their size rather than their signs; `lab_direct_route` (two rules), residual 0.2076 against a registered 0.20; its randomised-sign control is inverted in the same way, the coin residuals having median 0.0157 against μ's 0.2076 — smaller by a factor 13.2 — and that run's own diagnostic records that the null as designed tests nothing, since for a coin $\sum(\log k)H_\varepsilon$ and $\tilde{r} - \mathfrak{S}N$ are separately near zero and their difference is small for a reason unconnected with the cancellation; `lab_positive_weights`, top-decile share near 35% against a registered 40%; `lab_direct_identity` and `lab_layer_decomposition`, each on a rule about how much mass sits below the truncation; `audit_nogo_scope`, whose rule TD3 asked one hypothesis to cover both $b = \mu * 1$ and $b = \mu * \log$ and it covers neither, which is what Note 20 says in words; and `audit_squarefull_admitted`, whose V3 asked the note after Proposition 5 to reproduce its own printed three-decimal figures and one of them was a double rounding, corrected above to what that run reads. Of the eight, `lab_direct_route`'s Y1 is the one that bears on a statement: it is the convergence $B_{\log}(K) + \mathfrak{S}(N) \to 0$ that Proposition 12 uses, and what its failure records is that the run does not exhibit the rate at its own N, not that the limit is false. The other seven change no statement above — each was checked — and the reason for recording all of them here is that a reader who does not open the files would otherwise take the runs for clean.